

IP1: UNITY PROTOTYPE 1

XR Study Whiteboard

Student	Course	Date	Prototype
_____	DECO2300	27 August 2026	Unity 6.3.21f1

Project pitch

XR Study Whiteboard is a classroom whiteboard for extended reality. A student or teacher can draw on a wall whiteboard and write notes on paper on a table. The user uses a controller ray or a right hand pinch, so the writing is direct and easy to understand.

Prototype 1 purpose

This is a horizontal prototype. It gives a clear impression of the complete classroom use without trying to solve every future feature. The main goal is to test whether users can find each surface, draw accurately, and understand the tool controls.

Testing objective

Assumption to test: people can use the same basic drawing action on the wall and the table paper, while understanding that the paper has its own Pencil, Eraser, and Clear Paper controls.

- Can the participant aim at the correct surface and draw a continuous horizontal line, vertical line, and circle?
- Does the table paper look correctly placed and oriented while the participant writes?
- Can the participant find and use Pencil, Eraser, Clear Paper, colour, and Clear Board controls?
- Do controller rays, hand pinch, movement, and the physical marker feel clear enough to use?

Method

Use a moderated, task-based usability test. One participant uses the prototype for about 5 minutes while the facilitator gives short instructions and watches. Ask the participant to think aloud after each task. Record success, time, errors, comments, and confidence from 1 to 5. Aim for at least five participants in class.

Success criteria

Area	Pass condition
Drawing quality	The line or circle is continuous, smooth, and free from dots, gaps, glitches, and stray marks.
Surface targeting	The participant draws on the intended wall board or the top of the table paper.
Tool use	The participant can select a tool and can clear the correct surface.
Orientation	The paper and writing appear the right way round, with no mirrored or back-face marks.
Overall use	The participant can complete the core tasks without the facilitator taking control.

Prototype description and requirements

The Unity prototype places the user in a small XR classroom with a wall whiteboard, student tables, paper notes, chairs, lighting, and simple classroom objects. It is designed to look complete enough for a first evaluation and to make the main tasks visible at a glance.

- Wall whiteboard with black, red, blue, and green marker colours.
- Marker and larger eraser modes, plus a Clear Board confirmation step.
- Table paper with a finer Pencil, a wider Eraser, and a Clear Paper action for that paper only.
- Controller trigger and right hand pinch drawing paths.
- World-space controls with current tool, colour, and input status feedback.
- Controller rays, floor teleportation, snap turn, table view points, and a grabbable physical marker.
- Interpolated strokes and filtered input to keep lines and circles continuous.

Prototype stack

Part	Implementation used in this prototype
Engine	Unity 6.3.21f1 with an Android and OpenXR target.
XR input	XR Interaction Toolkit, OpenXR, XR Hands, and the Input System.
Drawing	CPU-updated textures with circular brush stamps and interpolation between input points.
Paper	Separate top writing surface and paper note texture, so notes do not appear on the back face.
UI	World-space UI controlled through the existing XR ray and hand interaction setup.

Testing setup

1. Open the XRStudyClassroom scene in Unity 6.3.21f1 and enter Play Mode, or start the Quest build.
2. Check that the right controller ray or hand tracking is visible and that the whiteboard and table paper are blank.
3. Keep the facilitator beside the participant with a timer and the data sheet. Do not explain a control unless the task allows help.
4. Reset the board and each paper after every participant. Record the device used and any tracking or comfort problem.

Testing process

Time	Facilitator says or does	Participant task
0:30	Explain that this tests the prototype, not the participant. Ask for think aloud feedback.	Look around and identify the wall whiteboard and one table paper.
1:00	Say: Draw one horizontal line, one vertical line, and one closed circle on the wall board.	Aim, hold the drawing input, and complete all three shapes.
1:00	Say: Move to the table and draw the same three shapes on the paper.	Use the table view or teleport controls, then write on the paper.
1:00	Ask the participant to change a tool, erase part of a mark, and clear only the paper.	Find TOOLS, use Pencil and Eraser, then use Clear Paper.
0:30	Ask what was easy, confusing, or uncomfortable. Record the final confidence score.	Describe the best and worst part and suggest one change.

Data collection sheet

Complete one row per participant. Use short notes so the results can be compared after class.

ID	Device	Wall board	Table paper	Controls	Main issue or quote
P1					
P2					
P3					
P4					
P5					

Current internal prototype check

- The wall whiteboard produced a clean horizontal line, vertical line, and closed circle during the internal editor test.
- The table paper produced the same three shapes. The strokes stayed continuous, smooth, and on the writing surface.
- The checks looked for dots, gaps, broken sections, random marks, sudden vertical or horizontal artefacts, and unwanted strokes. None were present after the fixes.
- Pencil, Eraser, Clear Paper, colour selection, and the Clear Board confirmation were checked. The table menu was moved so it did not overlap the writing area.
- The isolated Unity validation run passed 1042 checks. This verifies project logic and compilation, not real headset comfort or tracking.

Iteration record

Issue found	Change made	Retest result
Fast movement could make separate dots or gaps.	Interpolated points between samples and filtered input before stamping the texture.	Lines and circles became continuous on both surfaces.
The paper could show a wrong face or mirrored writing.	Used a top-only writing surface with corrected orientation and UV mapping.	Paper writing stayed on the top face and read the right way round.
The paper tools could compete with the writing area.	Repositioned the floating menu and kept the controls readable.	Pencil, Eraser, and Clear Paper were reachable without covering the paper.

Limitations and next steps

- Run the planned class test with at least five participants and use the data sheet to compare errors and confidence.
- Test controller tracking, hand pinch, haptics, locomotion comfort, and performance on a real Meta Quest headset.
- The prototype does not yet save notes or recognise handwriting. These are later features, not required for this first test.
- Use the test results to tune input thresholds, UI labels, menu placement, and the balance between the wall and paper workflows.

Originality and source note

This report describes the XR Study Whiteboard concept and its Unity prototype. AI-assisted support was used for code organisation, XR setup, debugging, and documentation. The student remains responsible for understanding, testing, explaining, and editing the final work. Project sources are recorded in Whiteboard/Documentation/ARCHITECTURE.md, CONTROLS.md, EDITOR_TESTING.md, IMPLEMENTATION_STATUS.md, and AI_USE.md.

Conclusion

Prototype 1 is horizontally complete for evaluation. It presents the classroom idea and supports the key actions on both drawing surfaces. The next evidence needed is participant testing on the target hardware, followed by documented design changes.